

Digital Twin, Agent and Artificial Intelligence Towards ION-2030

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Abstract This paper explores trends in Artificial Intelligence, Network Digital Twin, and Agents aimed at enhancing the operations of optical networks. Following the ITU-T ION-2030 vision, it also addresses the standardization gaps that remain. ©2026 The Author(s)

Introduction

ITU-T SG15 has recently published ITU-T GSTR-ION-2030 [1] as a strategic framework to guide the evolution of international optical networks.

Figure 1, which is a copy of Figure 1 of ITU-T GSTR-ION-2030 [1], summarizes the key characteristics and features of ION-2030.

ION-2030 divides the trends for Artificial

Intelligence (AI) into AI for ION-2030 and ION-2030 for AI. The former trends identify how AI can enhance the ION-2030 operations while the latter trends identify how ION-2030 can enhance the performances of AI applications by providing high bandwidth, low latency and high resilient connectivity services.

Figure 2, which is a copy of Figure 3 of ITU-T GSTR-ION-2030 [1], summarizes the AI-era optical network (AI-ON) architecture with ubiquitous AI covering both trends for AI and ION-2030.

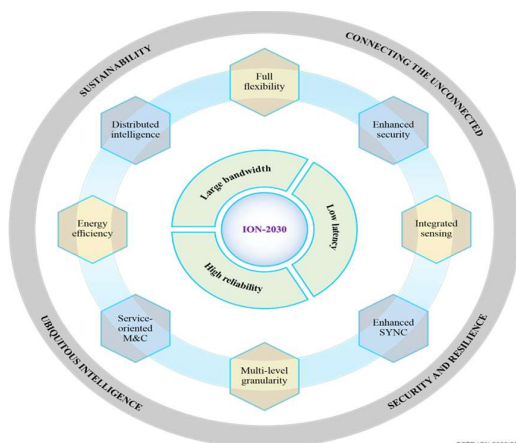


Fig. 1: The key characteristics and features for ION-2030.

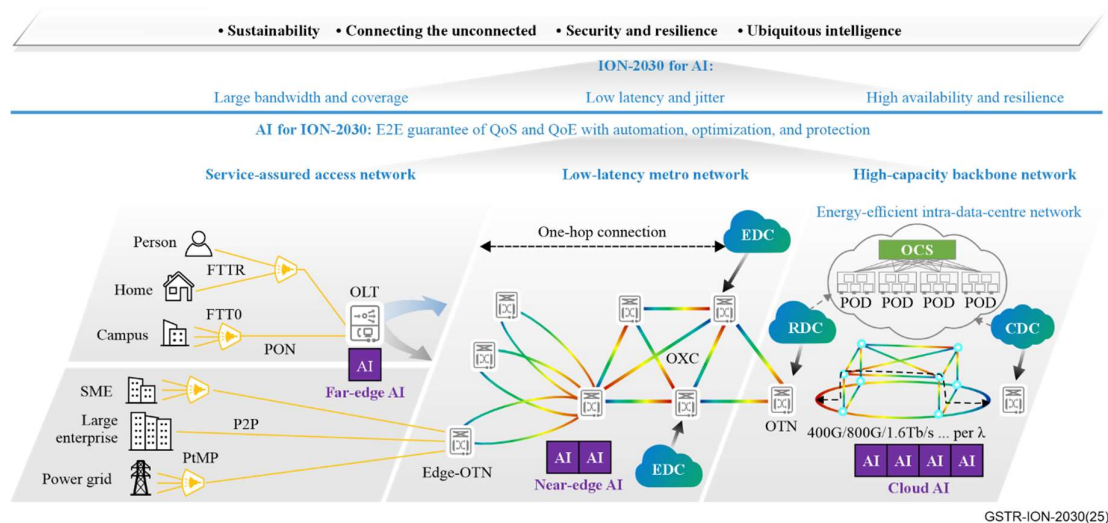


Fig. 2: AI-era optical network (AI-ON) architecture with ubiquitous AI.

Figure 3, which is a copy of Figure 4 of ITU-T GSTR-ION-2030 [1], summarizes the architecture of AI for ION-2030 which is composed by intelligent network elements, AI-native management and control (MC) systems, and AI-driven service platforms.

In particular, integrated sensing distributed in the network elements (NEs) provides to the MC system accurate and timely information about the state of the network. The network digital twin (NDT), based on this accurate information, can perform simulation and validation of planned deployment, optimization or maintenance operations.

The Network Optimization Agent monitors the network performance metrics (e.g., bandwidth utilization, latency, and throughput) to dynamically adjust routing policies and resource allocation based on real-time traffic analysis. It also proactively identifies bottlenecks and implements optimization measures to ensure optimal network performance.

The Fault Tracing Agent collects and correlates alarms from network elements to rapidly perform root cause analysis. It traces fault propagation paths across the network topology to facilitate efficient troubleshooting and automates

detect quality degradation before subscriber impact and it generates quality reports and trend analysis to support capacity planning and optimization decisions.

This paper describes some of the enhanced operations which can be enabled by adding an intelligent plane to the optical network elements and the MC system and it highlights the standardization gaps that should be addressed by on-going and future standardization work.

Enhanced operations in ION-2030

Adding an intelligent plane to optical network elements and MC systems would allow enhancing the operations of ION-2030.

This section describes some of the possible enhancements that can be achieved.

Knowledge-centric ontology would enable advanced AI-driven operation and maintenance (O&M) applications, thereby elevating network automation to a new level. AI agents, as discussed in the previous section, demonstrate strong capabilities in autonomous problem detection and resolution, addressing many limitations of traditional automation approaches.

However, model hallucination remains a critical risk in O&M applications. Addressing this

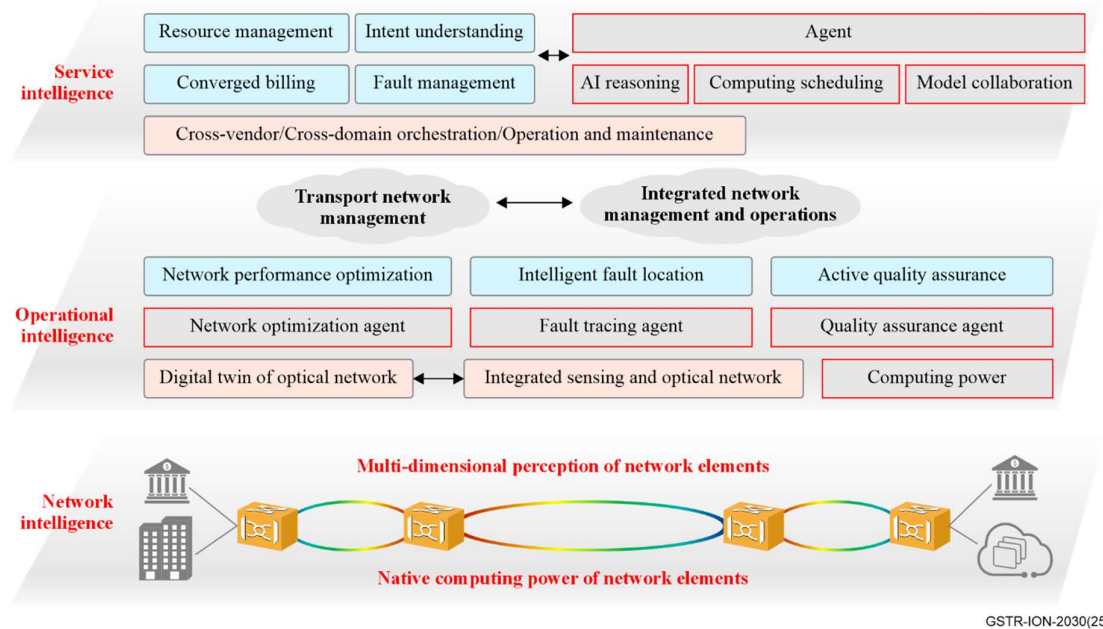


Fig. 3: AI-native optical network architecture.

fault resolution workflows to minimize Mean Time to Repair (MTTR) to enhance network reliability.

The Quality Assurance Agent continuously evaluates Quality of Service (QoS) and Quality of Experience (QoE) metrics against defined Service Level Agreement (SLA) thresholds. It conducts proactive end-to-end service probing to

challenge fundamentally depends on robust data governance rather than model optimization alone. A knowledge-centric ontology consolidates heterogeneous data sources and previously siloed information into a unified knowledge graph, providing a single source of truth.

This unified knowledge foundation facilitates

more reliable AI-driven inference by agents and, more importantly, helps mitigate hallucination by grounding model outputs in structured, verifiable knowledge.

Fibre sensing would allow detecting fibre quality deterioration, co-cable and co-trenching, identifying the risks before the fibre actually breaks and reducing the number of site visits for fault location after the fibre has broken.

The intelligent plane needs to monitor millions of environmental information (e.g., temperature distribution or vibration and acoustic disturbances along the fibre) and perform minute-level inference of risk locations.

During fault conditions, it can also use a Convolutional Neural Network (CNN) model for detecting the fibre fault location point.

Performance simulation models would allow improving the transmission performances by collecting thousands of optical parameters, including full data for Optical Amplifiers (OA), Wavelength Selective Switches (WSS), optical multiplexers, optical cross connects (OXC), and billions of samples, and performing multi-round of iterative simulations and continuous correction.

Intelligent dynamic energy saving would allow reducing the power consumption of the Optical Line Terminal (OLT) boards during low traffic or no traffic conditions.

The intelligent plane needs to perform a real-time analysis of the traffic pattern to identify the busy/idle states of the connected devices and be ready to quickly wake them up when there is traffic demand.

Intelligent home broadband troubleshooting would allow resolving failures in home broadband reducing the need for manual home visits.

The intelligent plane needs to perform an automatic closed-loop resolution of user mis-configuration such as Wi-Fi mis-configuration. Natural language interaction would assist the Network Operation Centres (NOC) engineers in quick decision-making and remote handling of the failures.

Intelligent Wi-Fi sensing would allow real-time accurate identification of various Wi-Fi interference sources and enable intelligent optimization algorithms for Wi-Fi power adjustment and signal transmission path optimization.

Capacity planning optimizations would allow planning the right network resources to meet the future demand to avoid over-provisioning or under-provisioning of network resources.

The intelligent plane analyses the traffic patterns to predict future demand and adjust capacity allocation accordingly.

Route optimization would allow improving the network utilization, reducing network

congestion and end-to-end latency.

The intelligent plane monitors the link utilization, latency, and failure conditions in real-time and, when needed, dynamically reroutes the traffic accordingly.

Preventive Maintenance Scheduling would allow reducing the MTTR and extend the equipment lifespan.

The intelligent plane performs predictive analytics on equipment health data to schedule maintenance before failures occur.

Standardization gaps

The emergence of AI technologies presents new challenges for standardization.

To address these, ITU-T SG15 is developing a new Technical Report on Enhanced Network Operations (GSTR-ENO), which defines the architectural framework and functional requirements to enhance optical transport network operations.

The document is scheduled for approval by July 2026. Its completion is expected to provide a clearer perspective on the interfaces between AI-related components and identify the standardization gaps to be addressed by future work within the ITU-T or other SDOs.

Conclusions

AI, specifically NDT and Agents, will play a critical role in enhancing optical network operations.

Integrating networks and AI, encompassing both “AI for networks” and “networks for AI” aspects, has become a strategic priority for global operators.

ITU-T GSTR-ION-2030 identifies AI-driven trends as key drivers for the evolution of optical networks.

While extensive research and development are currently focused on NDT and agent algorithms, the question of standardization remains open.

Consequently, ITU-T SG15 is actively working to identify and address existing gaps from a standardization perspective.

References

- [1] ITU-T Technical Report GSTR-ION-2030, "Technical report on international optical networks towards 2030 and beyond," Oct. 2025, <https://www.itu.int/epublications/publication/itu-t-gstr-ion-2030-2025-10-technical-report-on-international-optical-networks-towards-2030-and-beyond>